

>10 % in the range of 12–15 Gy. There was consistent agreement on low risk of RION for $D_{max} \leq 10$ Gy, and one major study indicated low risk for $D_{max} \leq 12$ Gy. BID treatment data and models indicate significantly larger tolerance at low dose per fraction. Risk increased significantly with age in studies examining the effect; more than doubling from 20–50 to 50–70 year groups. Data on other clinical factors such as chemotherapy, diabetes mellitus and hypertension were inconsistent. Linear quadratic models were inadequate for connecting RION dose thresholds for conventional fractionation to SRS fractionations. Models based on the RET or Neuret showed better agreement.

Conclusions: Collaborative research studies are needed to quantify IMRT dose-volumes safely tolerated above 55 Gy and to guide hypo-fractionated treatment protocols. Improving consistency within and among institutions in defining the optic nerves and chiasm is important for accurate determination of dose thresholds and dose-volume effects.

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2485 Cisplatin 30 Mg/M²/Week Decreases Toxicity without Sacrificing Efficacy in Patients Treated with Concomitant Chemoradiotherapy for Locally Advanced Squamous Cell Carcinoma of the Head and Neck

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Purpose/Objective(s): Toxicity is considerable with the standard chemotherapy regimen of cisplatin 100mg/m² every 3 weeks during radiotherapy (RT). We hypothesized that changing the chemotherapy regimen to a weekly dose of cisplatin 30 mg/m² would decrease toxicity without compromising efficacy in patients with locally advanced squamous cell carcinoma (SCC) of the head and neck.

Materials/Methods: The study population included 121 patients with AJCC Stage II (3%), Stage III (13%), or Stage IV (84%) SCC of the oropharynx (70%), hypopharynx (20%), or larynx (10%) treated from June 2000 to December 2006 at the University of Florida with altered fractionation RT and concomitant cisplatin given as a once-weekly infusion of 30mg/m². The median RT dose was 72.4 Gy given with either twice-daily fractionation (45%) or the concomitant boost (55%) technique. IMRT was used in each patient treated with the concomitant boost technique. Our endpoints were: analysis of acute and late toxicities, local-regional control (LRC), disease-free survival (DFS), cause-specific survival (CSS), distant metastases (DM), and overall survival (OS).

Results: The minimum follow-up for all patients was 2 years. Seventy-nine percent of patients were able to complete at least 6 cycles of chemotherapy. Ninety-five percent of patients received RT to at least 72 Gy. The 5-year actuarial outcomes were: LRC, 79%; CSS, 76%; DM, 12%; and OS, 59%.

Our toxicity rates (listed below) were lower than those reported for RTOG 9914 and 0129^{1,2}:

G-tube rate at the end of treatment: 52% (vs. 83% and 64%)^{1,2}

G-tube rate at 1 year: 3% (vs. 41% and 30%)^{1,2}

CTC Grade 4 late complications: 6% (vs. 16% and 20%)^{1,2}

CTC Grade 5 (fatal) toxicity: 2% (the same as seen in RTOG studies)^{1,2}

Conclusions: Our data suggests that, compared to more intensive regimens, weekly cisplatin at 30 mg/m² decreases toxicity without compromising tumor control in patients who receive concomitant chemo-RT for SCC of the head and neck. These results need to be confirmed in prospective studies.

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2486 Plaque Radiotherapy for Juxtapapillary Choroidal Melanoma: Visual and Systemic Outcomes

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Purpose/Objective(s): Management of choroidal melanoma adjacent to optic disc (juxtapapillary) is challenging due to the posterior tumor location, difficulty in localization and complexity of radiation field design. Various methods used to treat these tumors have included enucleation, proton beam radiotherapy, plaque radiotherapy, stereotactic radiotherapy and transpupillary thermotherapy (TTT). The Collaborative Ocular Melanoma Study (COMS) excluded treatment of lesions within 2 mm of the optic disc. We report the outcomes of treatment of juxtapapillary melanoma with plaque radiotherapy.

Materials/Methods: This was a retrospective, non-comparative review of 520 consecutive cases of juxtapapillary choroidal melanoma treated with custom designed plaque radiotherapy. The variables taken for analysis included the final visual acuity, radiation complications, recurrence of tumor, need for enucleation, systemic metastasis and mortality.

Results: The median age at presentation was 51 years (range, 15 – 93 years). The most common presenting symptoms were reduced visual acuity (51%), photopsia (10%) and visual field defect (13%). Optic disc overhang by tumor was absent in 389 eyes (75%), 1% to 50% overhang in 83 (16%) and 51% to 100% overhang in 48 (9%). The quadrant location of the main tumor was superior in 146 (28%) eyes, inferior in 114 (22%), nasal in 137 (26%) and temporal in 123 (24%). The median tumor basal diameter was 9.5 mm (range, 1.5–20 mm) and thickness 3.5 mm (range, 1.0 – 14.8 mm). The radioactive isotopes used were Iodine ¹²⁵ in 504 eyes (97%), cobalt ⁶⁰ in 8 (2%) and iridium ¹⁹² in 8 (2%). The plaque shape was notched in 471 eyes (91%), round in 42 eyes (8%) and rectangular in 7 (1%). The median radiation dose to the apex of the tumor was 8000 cGy, and to the base was 28,500 cGy. Kaplan-Meier analysis at 2, 5 and 10 years showed poor visual acuity ($\leq 20/200$) in 21%, 55%, and 84%; visual loss of more than 5 acuity